Derandomized Non-Abelian Homomorphism Testing in Low Soundness Regime

Tushant Mittal

Joint work with Sourya Roy, University of Iowa

Homomorphism Testing

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Representation Theory

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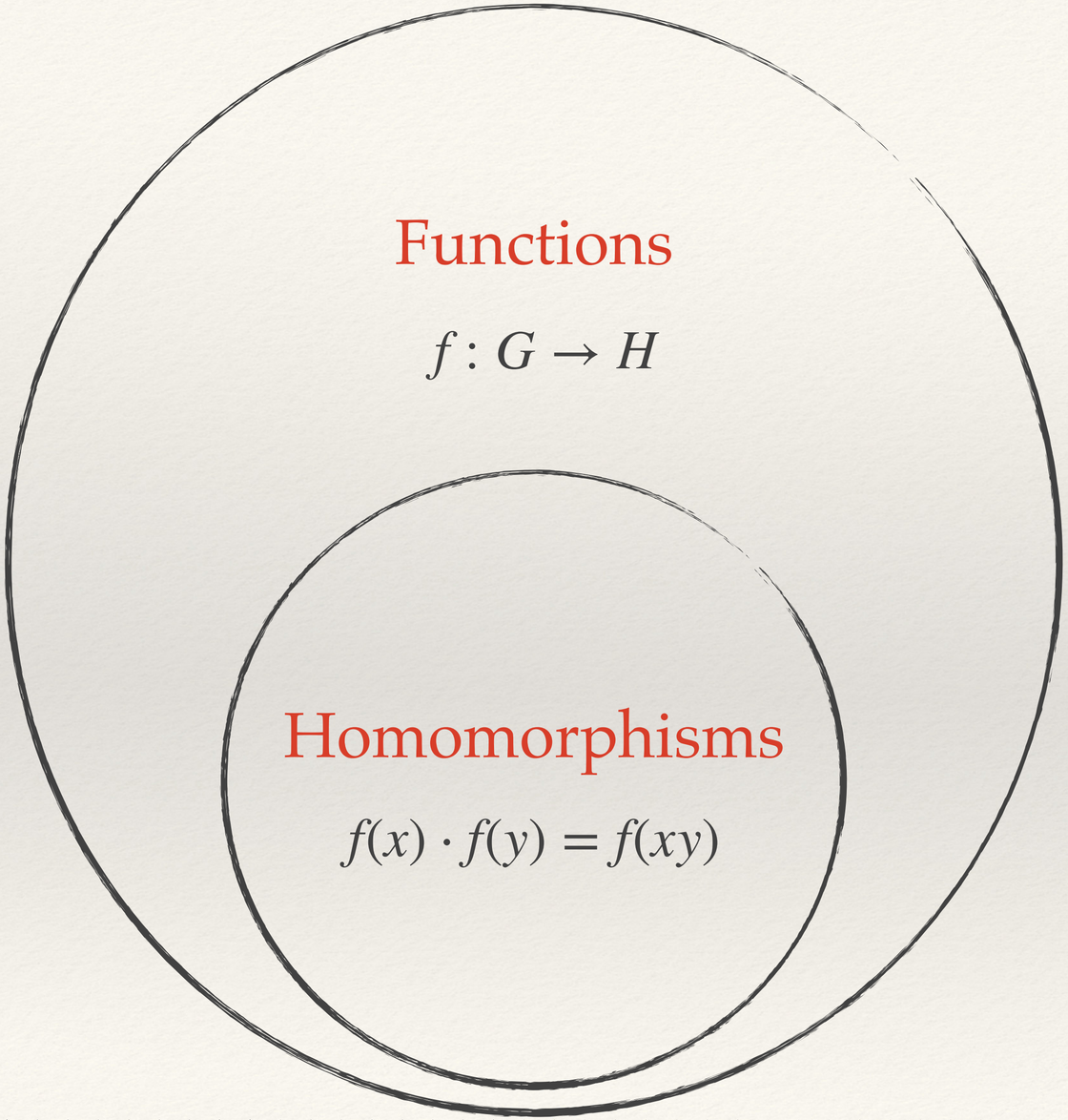
Derandomization Technique

Functions

$f : G \to H$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Homomorphisms
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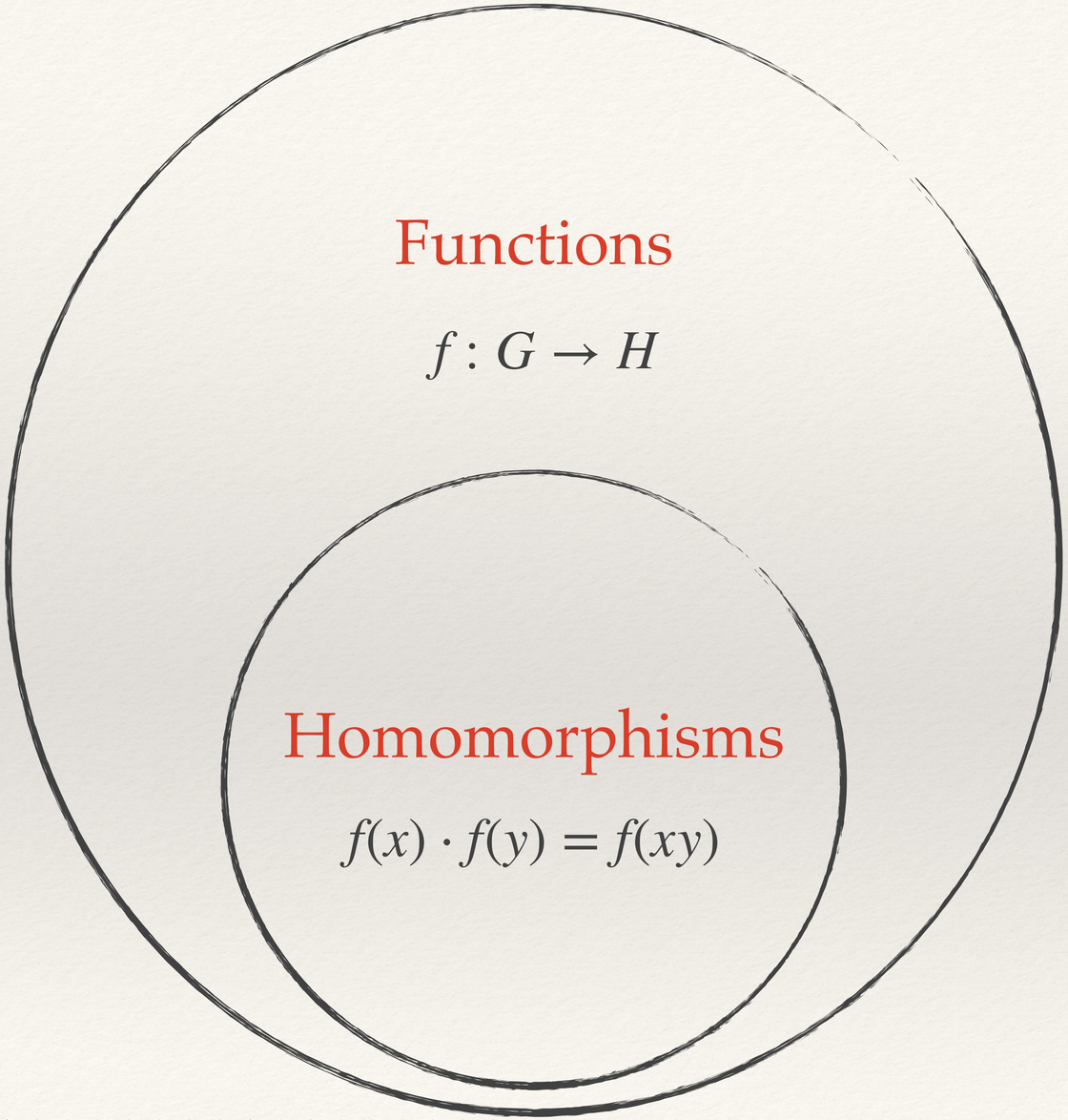
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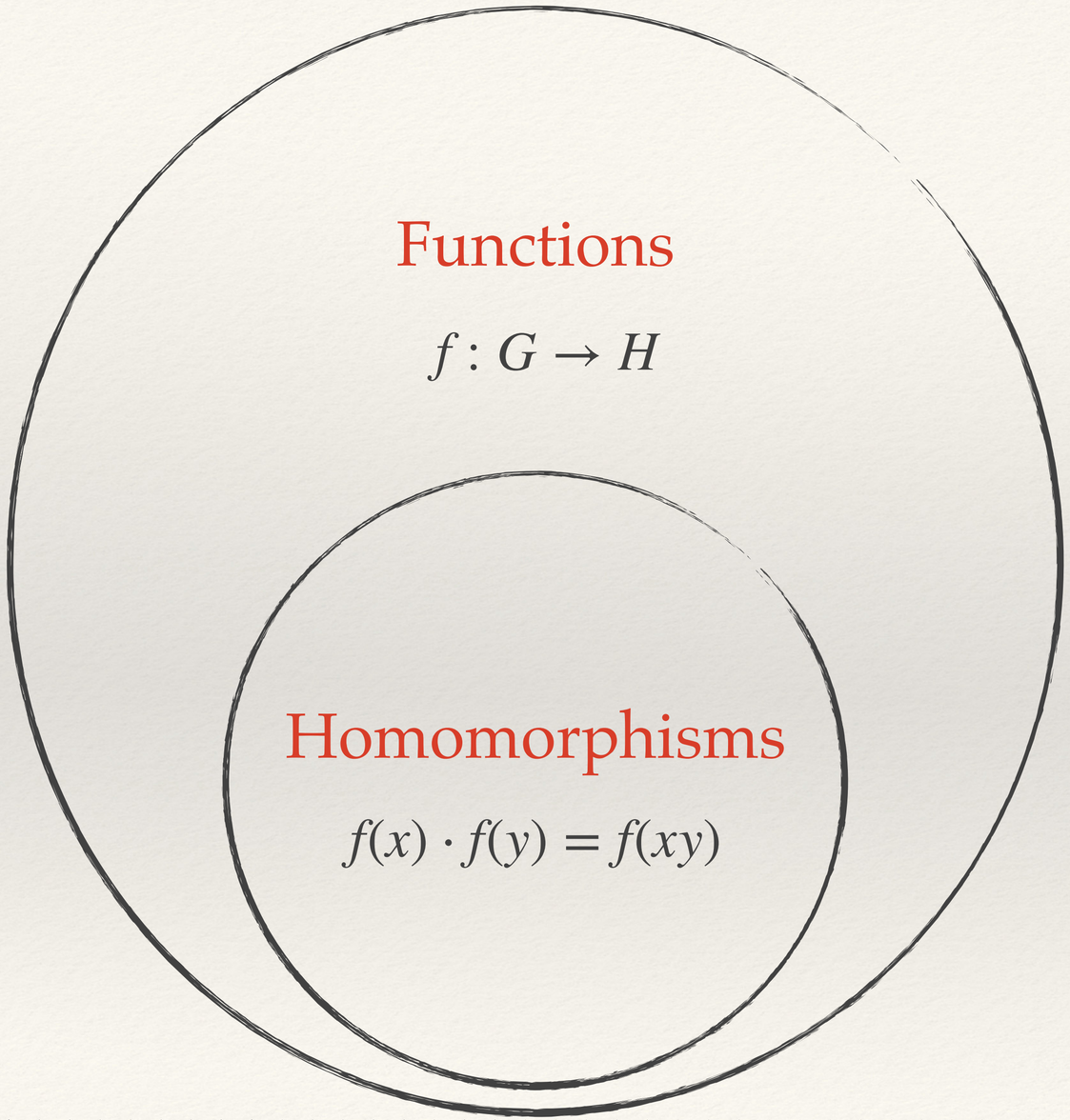

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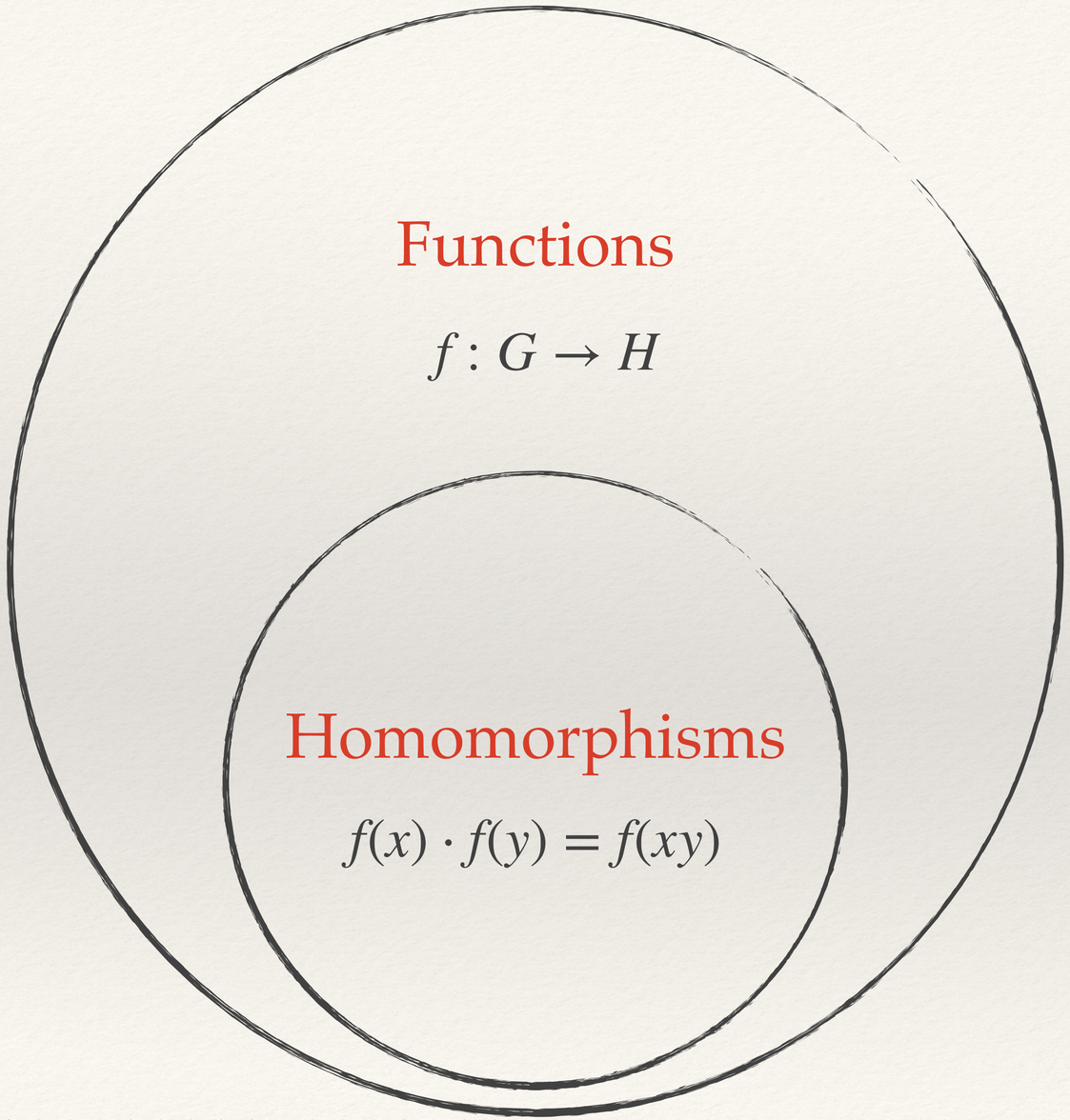

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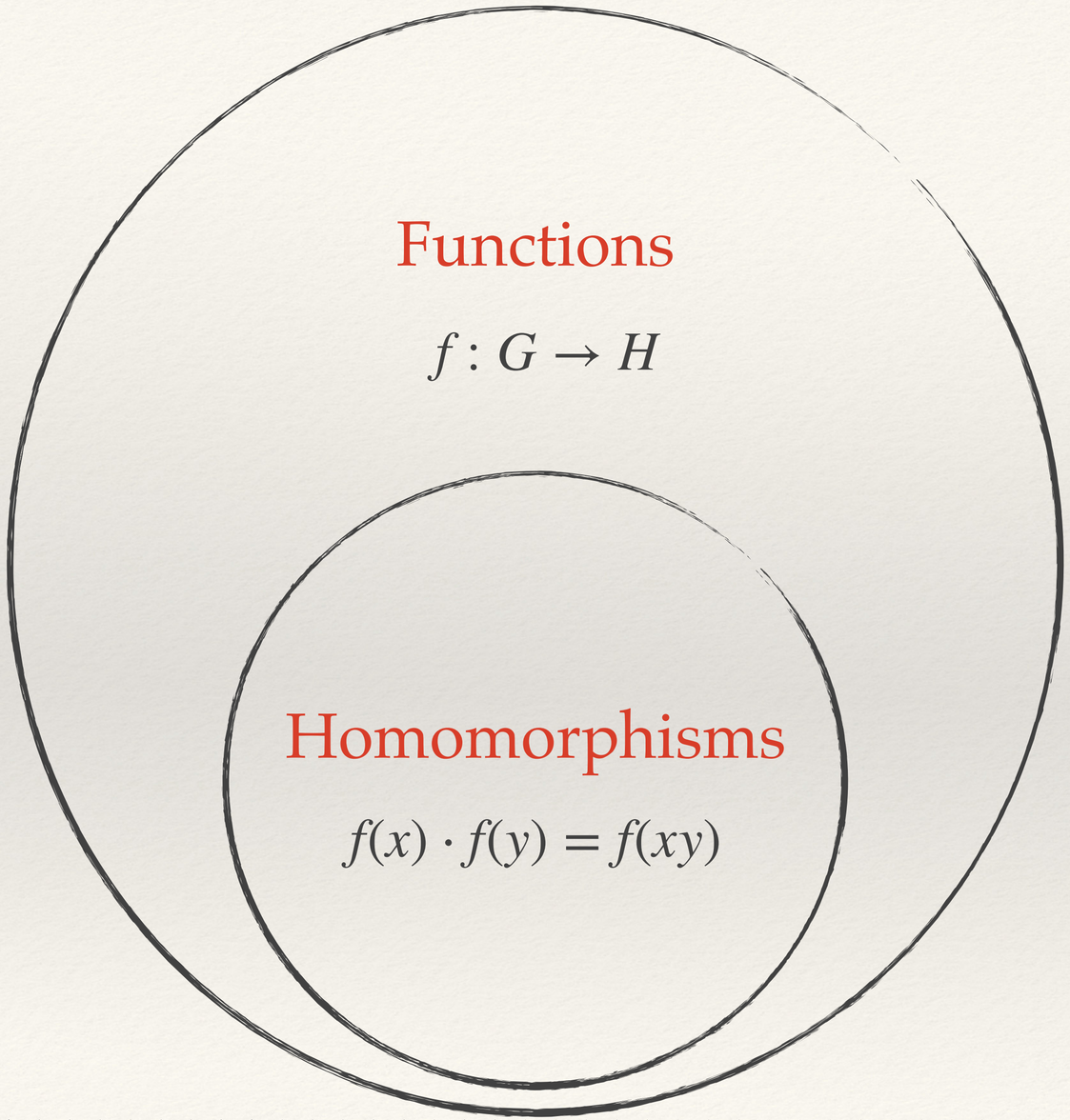

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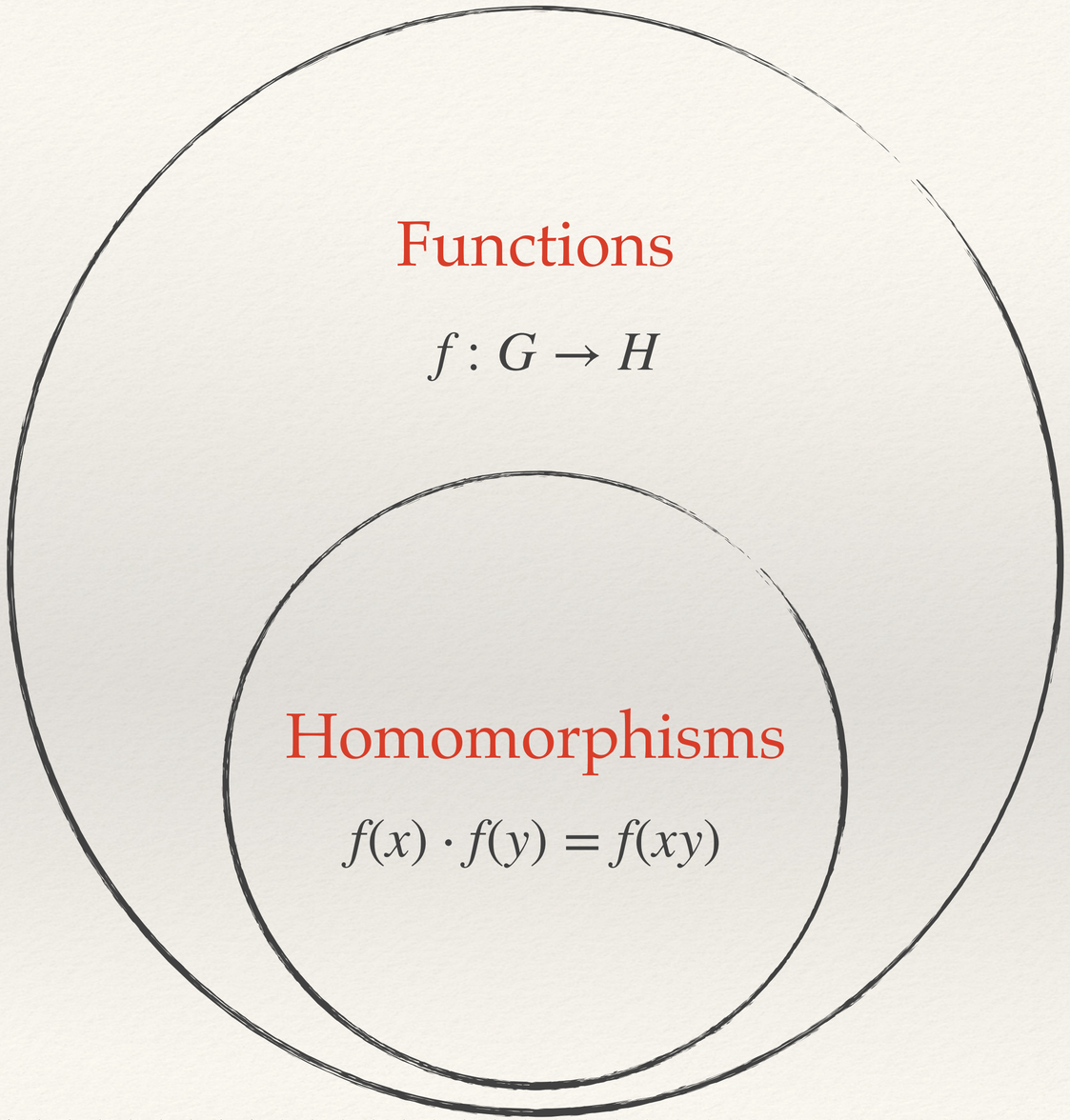

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Testing

For a constant fraction of pairs,

$f(x) \cdot f(y) = f(xy)$.

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  - $\delta := \Pr_{x,y \sim G} \left[f(x) f(y) \;=\; f(xy) \right]$

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- Key Question — What does δ or (δ_S) say about f?

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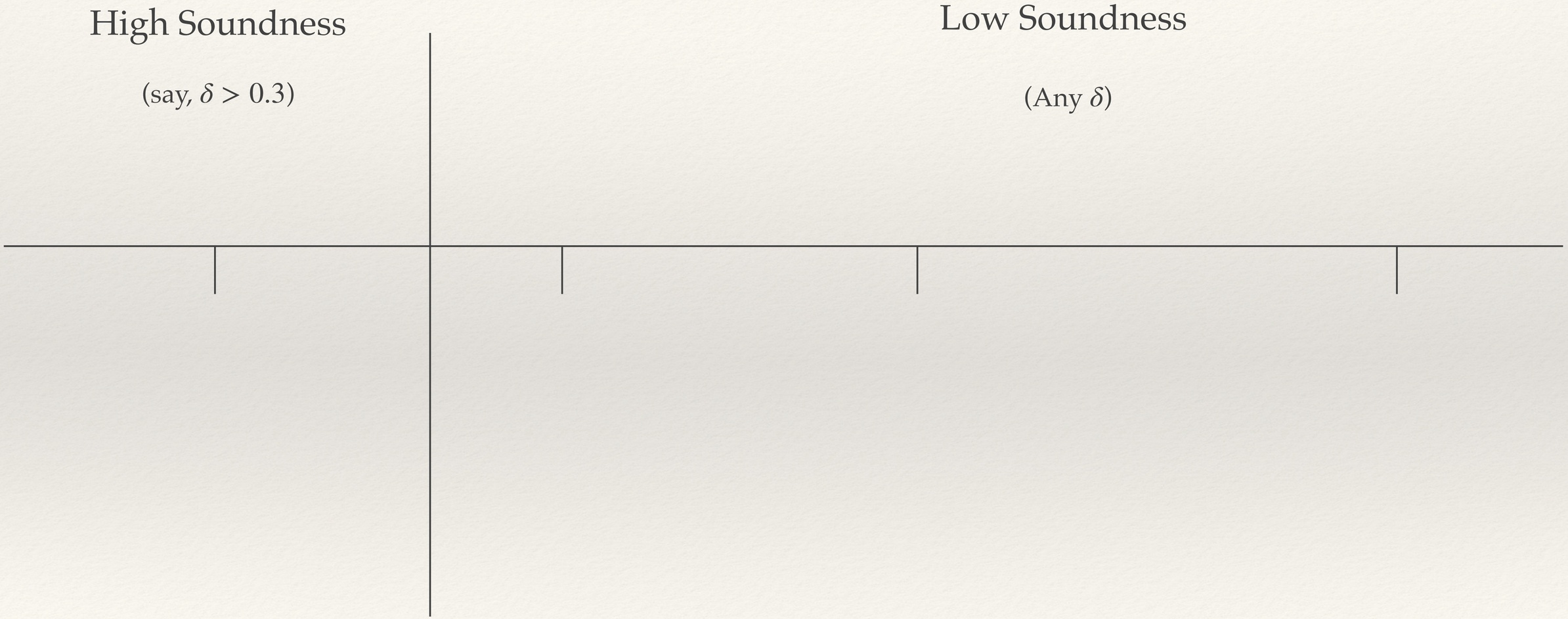
  - f is $(1 - \delta)$-close to a homomorphism in the Hamming metric.

Prior Work



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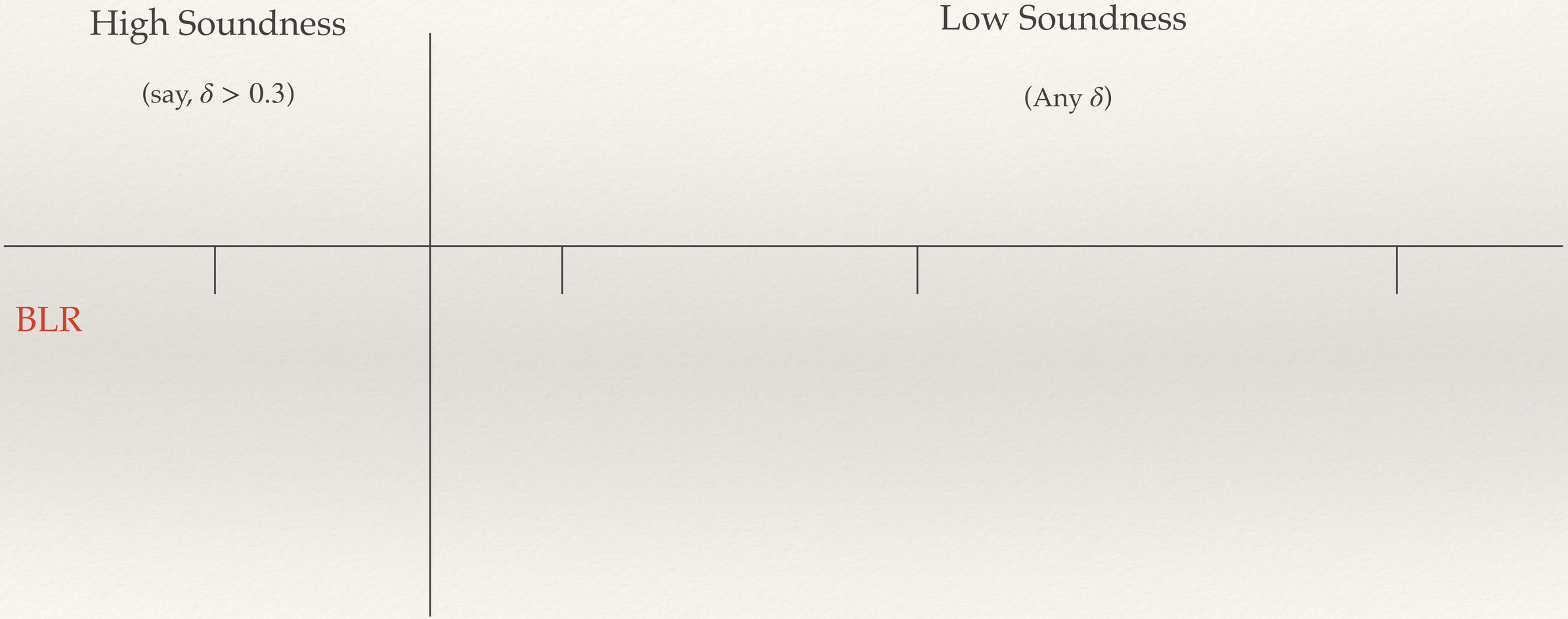

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(say, $\delta > 0.3$)

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(Any δ)

BLR

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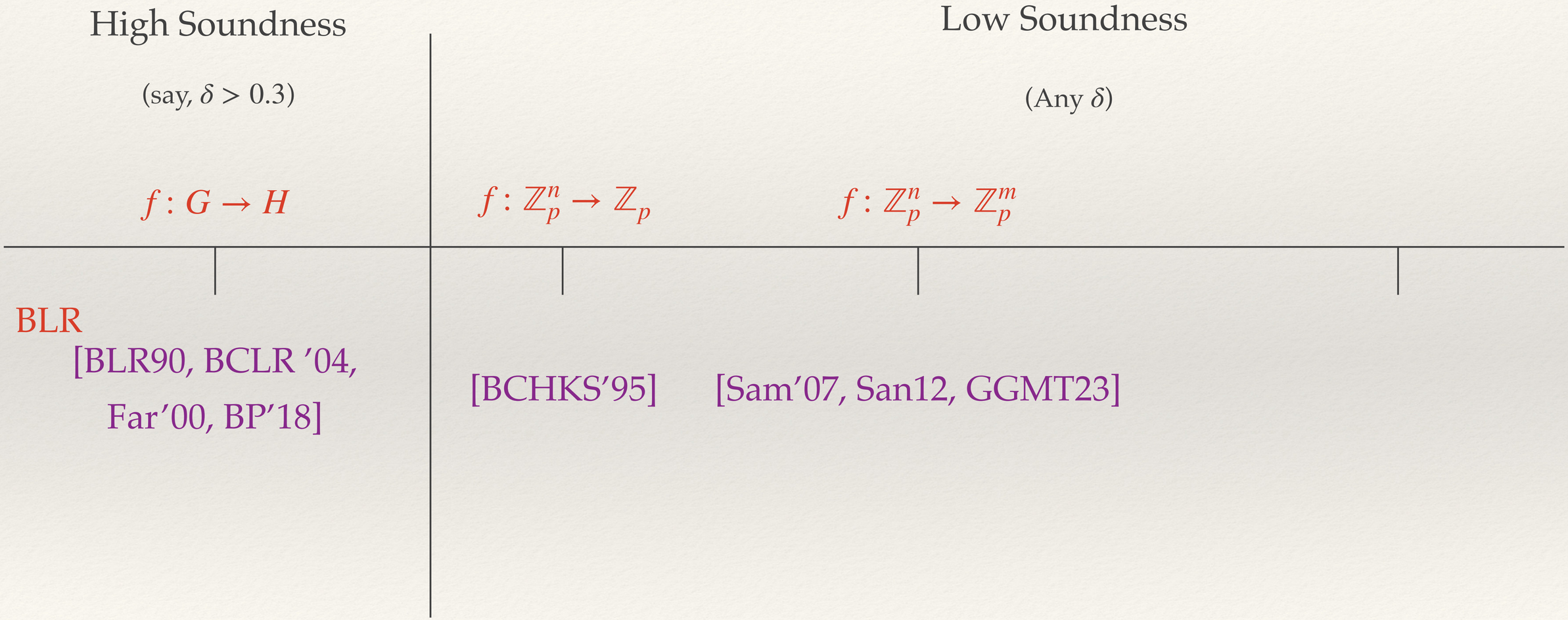
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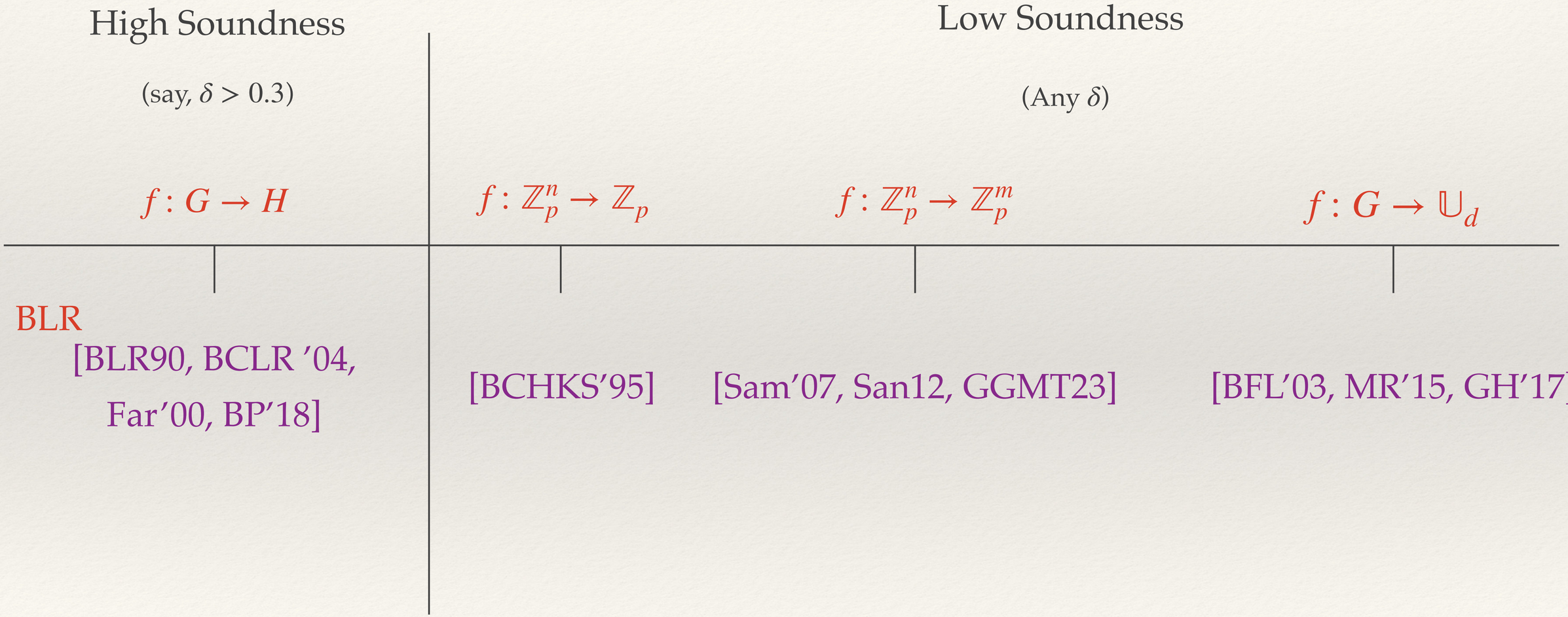
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Derandomized BLR [SW'04]	[BSVW '03]		[Our Result]

Representation Theory

(The study of homomorphisms $f : G \to \mathbb{U}_d$)

Gambling is nothing but the applied representation theory of the symmetric groups. — I. V. Cherednik

Fourier basis

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  - $\{\chi_T\}_T$ forms an orthogonal basis for functions on hypercube, $L^2(\mathbb{Z}_2^n)$.

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$$\rho(\sigma) = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \mathbb{I}[\sigma(3) = 2] \\ \cdot & \cdot & \cdot \end{bmatrix}$$

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  - Property 1 — $\rho_{i,j}$ comes from an irrep.
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  - Property 3 —Every homomorphism $\varphi : G \to \mathbb{U}_n$, splits as $\varphi = \bigoplus_\rho \rho$.

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- Random sets of size $O(\log |G| / \varepsilon^2)$ are small-bias.
  - Better constructions than random ones available!

Non-Abelian Testing

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Linearity Testing

$f : \mathbb{Z}_2^n \to \mathbb{Z}_2$ passes the

BLR test with probability δ

Close to a true homomorphism

Large Fourier Coefficient

Non-Abelian Groups

$f : G \to \mathbb{U}_1$ passes the BLR test with probability δ

Correlated with a "scaled homomorphism"

Small irrep with large mass

Non-Abelian Groups

[Gowers-Hatami'17]

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$\eta = \delta^2/2$

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Small irrep with large mass

$\|\hat{f}(\rho)\|_{\mathrm{HS}}^2 \geq \eta$ for some irrep ρ of dimension at most $\frac{1}{\eta}$.

Main Theorem

$f : G \to \mathbb{U}_1$ passes the derandomized BLR test with probability δ_S.

$$\eta = \frac{\delta_S^2 - \varepsilon}{2}$$

Correlated with a "scaled homomorphism"

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Main Theorem

$f : G \to \mathbb{U}_d$ passes the derandomized BLR test with probability δ_S.

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Correlated with a "scaled homomorphism"

$$\langle f, \pi \rangle \geq \eta^2 \cdot d\,, \text{ where } \pi = U\rho V^*.$$

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Proof Ideas

Small *algebra norm* functions are “fooled” by small-bias sets

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New Degree-2 Expander Mixing Lemma (EML) for Cayley Graphs

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→ New Degree-2 Expander Mixing Lemma (EML) for Cayley Graphs

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[GH'17, MR'15]

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New Degree-2 Expander Mixing Lemma (EML) for Cayley Graphs

[GH'17, MR'15]

Derandomized BLR Test

Derandomized "BNP" Lemma

[BNP'05, Gowers'06]

Rest of the talk

Small *algebra norm* functions are "fooled" by small-bias sets

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- Algebra norm - $\|h\|_A = \|T_h\|_{\mathrm{tr}}$ where $T_h(x,y) = h(xy^{-1})$.

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$$\|f\|_{U^2}^4 = \Sigma_\chi \hat{f}(\chi)^4 \geq \delta^2 \implies \max_\chi |\hat{f}(\chi)| \geq \delta \iff \min_\chi \|f - \chi\|_{\text{Ham}} \leq 1 - \delta.$$

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  - Equivalent to S being ε-biased.

Conclusion

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  - Mixing Lemmas — New ones and other uses.

Thank You!

Questions?